A1 production kernel manifest – full runtime scalar-slice from JSON (v1.5)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

first issued 2026-06-23 · this version issued 2026-06-23 · v1.5

1. Purpose and scope

Certifying the scalar Brazovskii slice as the real N-001 kernel requires that the verifier evaluate the ORIGINAL `bloch_matrix_linear` under the EXACT runtime settings, not checker-chosen defaults. The function's scalar diagonal depends on `laplacian_mode`, `bcc_mix_epsilon`, `a_bcc`, `family_masses`, `k_lock` (function DEFAULT 0.15) and `eta_shell`; v1.4 hardcoded two of these. v1.5 reads all of them from the config and passes them verbatim. Verification: `codes/foundations/a1_kernel_checks.py` v1.5 (14/14 PASS); solver sources `codes/foundations/n001_solver/`; config `claims/A1-PRODUCTION-KERNEL-MANIFEST/canonical_n001_kernel.json`.

2. The runtime scalar slice

The config stores the full param set passed to the original function:

$$\texttt{scalar\_slice} = \{\texttt{laplacian\_mode}{=}\text{spectral},\ \texttt{bcc\_mix\_epsilon}{=}0,\ \texttt{a\_bcc}{=}1,\ \texttt{family\_masses}{=}\mathbf{0},\ \texttt{k\_lock}{=}0,\ \texttt{eta\_shell}{=}0\}. \tag{1}$$

Under these, the ORIGINAL $\texttt{bloch\_matrix\_linear}(\{|k|,0,0\},\{r,Z,Y,q_0,\texttt{**scalar\_slice}\})_{00}$ equals the stored $r_{\text{zero}} + Z|k|^2 + Y|k|^4$ to 8.9×10^{-16}. Each setting is load-bearing (reverting it shifts the ORIGINAL diagonal): `k_lock`=0.15 (the function default) $\to$ 0.100 via $k_{\text{lock}}(I - P_0)_{00} = 0.15 \cdot \frac{2}{3}$; `eta_shell`=0.1 $\to$ 0.231; `mixed_bcc`(ϵ=0.5) $\to$ 1.997; `family_masses` $\to$ 0.050. Thus fixing them in the CONFIG (read by the checker) rather than in the checker is essential to verifying the runtime symbol.

3. Schema completeness and provenance

The verifier asserts the config carries all required keys (the six physical fields, three tolerances, the seven `scalar_slice` keys, the `origin_n001_solver` hash map) BEFORE using them, so a stale/v1.2 config fails the named claim `config_schema_v15_complete` rather than raising `KeyError`. The three original solver sources are vendored byte-identical under `n001_solver/` with full sha256 pinned and checked (`continuation_mu2_v25.py 6db0393a..`, `bloch_linearization.py 337bd6de..`, `math56_constants.py 6c808597..`). The withdrawn v1.3 mimic `actual_n001_pde_backend.py` is excluded from the v1.5 bundle and no config key references it.

4. Status: gates pass, certification pending

Under the runtime scalar slice the canonical N-001 config clears both manifests ($\delta_Z = \delta_r = 0$, $\delta_m = 4.3 \times 10^{-18}$, `mu2_shell` > 0) AND the ORIGINAL solver symbol equals the stored kernel. The script reports "numerical gates pass; operator certification pending": T1 OPEN, pending the operator's independent run.

5. Devil's-advocate

(α) *"Checker still chooses settings."* DISMISSED — all scalar-slice settings are read from the JSON and shown load-bearing; the checker injects none. (β) *"Stale config slips through."* DISMISSED — the schema-completeness claim fails loudly on any missing key. (γ) *"Scalar slice $\neq$ full operator."* VALID and explicitly scoped — A1/A2/A3 certify the pure scalar Brazovskii kernel; the anisotropic / locked / shell-biased operator is the mainline production-Hessian question. Quantitative sanity check: runtime symbol residual 8.9×10^{-16}; $k_{\mathrm{lock}} = 0.15$ shift $0.100 = 0.15 \cdot \frac{2}{3}$ (exact).

6. Result footer

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Result ID:              A1-PRODUCTION-KERNEL-MANIFEST (OPEN, v1.5)
Precise statement:      Verifier asserts config schema complete, reads 6 physical
                        fields + the 7-key scalar_slice from canonical_n001_kernel
                        .json, imports the ORIGINAL solver and evaluates
                        bloch_matrix_linear(|k|, {r,Z,Y,q0, **scalar_slice}) =
                        r_zero+Z|k|^2+Y|k|^4 (8.9e-16) -- the runtime symbol; each
                        scalar_slice setting shown load-bearing (k_lock=0.15->0.1,
                        eta_shell=0.1->0.231, mixed_bcc->2.0, family->0.05); 3
                        source sha256 pinned.
Scope:                  pure-Brazovskii scalar slice; CONSISTENCY + ANALYTIC-BRANCH.
Dependencies:           A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Bundle:                 canonical_n001_kernel.json + continuation_mu2_v25.py +
                        bloch_linearization.py + math56_constants.py + a1_kernel_checks.py.
Evidence grade:         ANALYTIC + EXECUTED (14/14) -- pending operator independent run.
Reproduction command:   python codes/foundations/a1_kernel_checks.py
Expected output:        14/14 PASS; schema complete; runtime symbol matches; all
                        scalar_slice settings load-bearing; sha256 pinned.
Tier:                   T1 OPEN (gates pass; certification pending).
Falsifier:              any scalar_slice setting != stored, eta_shell != 0, a source
                        -hash mismatch, a stored-field edit breaking a delta, or
                        kinetic_coefficients not reproducing (r_zero,Z).
No-overclaim:           scalar slice only (full anisotropic operator excluded); gates
                        passing != certification; not yet A2/A3-citable.
Next required action:   Operator independently runs the bundle (14/14) and certifies.
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